

```
// SPDX-License-Identifier: MIT
```

```
pragma solidity ^0.8.18;
```

```
/// @title Simple Bank Account
```

```
contract BankAccount {
```

```
    address public owner;
```

```
    event Deposit(address indexed from, uint256 amount);
```

```
    event Withdraw(address indexed to, uint256 amount);
```

```
    modifier onlyOwner() {
```

```
        require(msg.sender == owner, "Only owner");
```

```
        _;
```

```
    }
```

```
    constructor() {
```

```
        owner = msg.sender;
```

```
    }
```

```
    receive() external payable {
```

```
        emit Deposit(msg.sender, msg.value);
```

```
    }
```

```
    fallback() external payable {
```

```
        if (msg.value > 0) emit Deposit(msg.sender, msg.value);
```

```
    }
```

```
    function deposit() external payable {
```

```
        require(msg.value > 0, "No Ether sent");
```

```
        emit Deposit(msg.sender, msg.value);
```

```
    }
```

```
    function withdraw(uint256 amount) external onlyOwner {
```

```
        require(amount > 0 && amount <= address(this).balance, "Invalid amount");
```

```
        (bool ok, ) = payable(owner).call{value: amount}("");
```

```
        require(ok, "Transfer failed");
```

```
        emit Withdraw(owner, amount);
```

```
}  
  
function getBalance() external view returns (uint256) {  
    return address(this).balance;  
}  
  
}
```